Data Science:

Design and develop statistical and machine learning models to analyze structured and unstructured data for predictive and prescriptive insights. Perform data cleaning, feature engineering, model validation, and performance optimization using programming languages such as Python or R and relevant ML frameworks. Collaborate with engineering and business teams to deploy models, interpret results, and support data-driven decision-making.

Statistics:

Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data to make informed decisions.

Inferential (statistics)

using a *sample* of data to draw conclusions about a larger population, and checking whether those conclusions are likely to be true using probability.

Inferential doesn't "prove" something 100% — it tells you how strong the evidence is, and how likely your result could happen by chance.

Descriptive Statistics

Descriptive statistics summarize and describe the main features of a dataset.
They do not make predictions or generalizations — they only explain what the data shows.

Types of Data

1) Categorical (Qualitative Data)

Non-numeric labels or groups.

Two main types:

- **Nominal** → No order(Summer, winter, rainy, autumn—this occurrence)
    - Gender
    - Color
    - Country
- **Ordinal** → Has order, but differences are not measurable
    - Small, Medium, Large
    - Education level
    - Customer rating (1–5 stars)

Example:
Region = East, West, South
Numerical (Quantitative Data)

Numeric values where arithmetic makes sense.
Numerical data is divided into:

A) Discrete

Countable values (usually integers)

- Number of customers
- Number of products sold
- Number of clicks

Example: 5, 10, 25

B) Continuous

Measured values (can take decimals)

- Height
- Weight
- Temperature
- Revenue

Example: 72.5 kg, 99.99 dollar

Correct Structure

Data Types

1.Categorical
  Nominal
  Ordinal

2. Numerical
  Discrete
  Continuous

Levels of Measurement

1)Nominal

Categories with **no order**

- Gender
- Country
- Product category

Only counts and percentages make sense.

2)Ordinal

Categories with **order**, but differences are not equal

- Customer rating (1–5 stars)
- Education level
- Satisfaction: Low, Medium, High

You can rank them, but you cannot assume equal spacing.

3)Interval

Numeric scale where:

- Differences between values are meaningful
- **No true zero** (zero does NOT mean absence)

Examples:

- Degree Celsius
- Degree Fahrenheit
- Calendar years

Important:

0°C does NOT mean “no temperature.”

You can:

- Add / subtract

  You cannot:
- Say 40°C is twice as hot as 20°C

4)Ratio

Numeric scale where:

- Differences are meaningful
- There IS a **true zero** (zero = absence of quantity)

Examples:

- Height
- Weight
- Revenue
- Age
- Kelvin temperature

Here, ratios make sense:

- 20 kg is twice 10 kg
- $200 is twice $100

Qualitative (Categorical Data)

Describes *qualities, attributes, or labels* — not measurable numerically.

Example:

- Eye color (Brown, Blue, Green)
- Gender
- Country
- Product category

Key properties:

- You **cannot meaningfully add or average** them.
- You summarize using:
    - Counts
    - Percentages
    - Mode
    - Bar charts

Quantitative (Numerical Data)
Definition:
Represents measurable quantities — numbers where arithmetic makes sense.
Example:

- Height (170 cm)
- Weight (65 kg)
- Salary ($75,000)
- Age (25 years)

Key properties:

- You **can add, subtract, multiply, average**
- You summarize using:
    - Mean
    - Median
    - Variance
    - Standard deviation
    - Correlation

Mean (Average)

Add all values and divide by number of observations.

Uses every value
Sensitive to outliers

Median (Middle Value)

Steps:

1. Sort the data
2. Pick the center value

Not affected much by outliers
Good for skewed data

If odd number of values → pick middle
If even → average of two middle values

Mode (Most Frequent Value)

The value that appears **most often**.

Works for both numerical and categorical data
Not affected by extreme values

Measures of Dispersion (Spread of Data)

While central tendency tells you *where the center is*, dispersion tells you **how spread out the data is around that center**.

Variance

Average of squared deviations from the mean.

Standard Deviation

Square root of variance.

Normal Distribution (Gaussian Distribution)

The **normal distribution** is a continuous probability distribution that is symmetric and bell-shaped. It is one of the most important concepts in statistics and machine learning.

Mean = Median = Mode

All three measures of central tendency are equal and located at the center.

Symmetry

Perfectly symmetric around the center.

Skewness
Skewness measures the **asymmetry of a distribution** around its mean.

It tells you whether the data is tilted left or right.

Types of Skewness

Zero Skewness (Symmetric Distribution)

Mean = Median = Mode

Positive Skewness (Right-Skewed)

Long tail on the right

Mean > Median > Mode

Negative Skewness (Left-Skewed)

Long tail on the left

Extreme low values pull mean downward

Mean < Median < Mode

Skewness is a property of the distribution.

Removing extreme outliers:

- **May reduce skewness**

Skewness = 0 → symmetric

Skewness > 0 → right skew

Skewness < 0 → left skew

Random Variable

A random variable is a function that assigns a numerical value to each possible outcome of a random experiment.